

CS3103 - Computer Networks Practice
Audit
Summary

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1 Course Overview

1.1 Module Information

- **Module:** CS3103 - Computer Networks Practice
- **Textbook:** *Computer Networking: A Top-Down Approach* (6/e or 7/e) by KUROSE and ROSS (Pearson).
- **Tools:** All lab experiments are performed on **GNS3**. Discussions, feedback, and announcements are on **Discord.com**.
- **Philosophy:** "Build, Configure and Observe the Protocols in Action, Design protocols and algorithms for Modern networks and Develop Applications with the Protocols".

1.2 Assessment Components (Approximate)

- **Practice - LAB** (30%): Lab Sheets/Questionnaires and Pre-lab Quizzes.
- **Practice - PROGRAMMING ASSIGNMENTS** (35%): Project/Assignments (one group-project and two or three individual-assignments).
- **UNDERSTANDING CONCEPTS** (35%): Written Test (Open Book) and Participation.

2 TCP/IP Networking: Review and Addressing

2.1 Address Resolution Protocol (ARP)

- **Motivation:** To deliver an IP packet to the next device within a LAN (hop-to-hop), the IP address must be translated to a physical (MAC) address.
- **Mechanism:**
 1. **ARP Request:** Sent as a **broadcast** on the LAN: "What is the MAC address of [Target IP]?".
 2. **ARP Reply:** The target machine responds with a **unicast** containing its MAC address.
- **Proxy ARP:** A router responds with its own MAC address to an ARP request for a device on a different, connected subnet, effectively handling the routing for the client.

2.2 IP Address Structure and Classes

- **Structure:** An IP address consists of a **netid** and a **hostid**, separated by a **net-mask** or **prefix length** (CIDR).
- **Special Addresses:**
 - **Network ID:** The first address in the network (host part all zeros).
 - **Direct Broadcast Address:** The last address in the network (host part all ones).
- **Classful Addressing (Obsolete):** Used fixed prefix lengths (/8, /16, /24). Was inefficient (e.g., millions of addresses wasted in Class A).
- **Classless Addressing (CIDR):** Uses **Variable Length Subnet Mask (VLSM)**, allowing any prefix length, to solve address depletion and enable **Supernetting** (route aggregation).

3 Dynamic Host Configuration (DHCP), Subnetting, and NAT

3.1 DHCP Protocol (Dynamic Host Configuration Protocol)

- **Function:** DHCP dynamically assigns an IP address (logical address) from a pool for a specific duration (lease).
- **Protocol:** Application Layer protocol using UDP (Server: Port 67, Client: Port 68).
- **Lease Process (DORA):**
 1. **DHCPDISCOVER** (Broadcast to find server)
 2. **DHCPOFFER** (Server proposes an address and lease)
 3. **DHCPREQUEST** (Client selects and requests offer)
 4. **DHCPACK** (Server finalizes the lease/settings)
- **Renewal:** Clients attempt to renew the lease at the **Renewal Timer** (50% of lease time).
- **DHCP Relay Agent:** A router that forwards broadcast DHCPDISCOVER messages (which cannot cross subnets) to a centralized DHCP server as a unicast message.

3.2 Network Address Translation (NAT)

- **Purpose:** Allows a private network (using non-globally-routeable IP addresses) to share a single public IP address for Internet access.

- **Private IP Ranges (RFC 1918):** 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16.
- **Mechanism (PAT):** The NAT router maintains a **translation table** (Five-column table) mapping internal (Private_IP:Private_Port) traffic to external (Public_IP:External_Port).

4 Virtual LANs (VLAN) and Wireless Networking

4.1 Virtual LANs (VLAN)

- **Concept:** A logical broadcast domain independent of physical location, grouping devices by function/department.
- **VLAN Tagging (IEEE 802.1Q):** Used on **trunk ports** (links carrying multiple VLANs) to carry the **VLAN ID (VID)** (12 bits) within the Ethernet frame header.
- **Implementation:**
 - **Static (Port-Based):** VLAN assigned based on switch port.
 - **Dynamic (MAC-Based):** VLAN assigned based on the device's MAC address (for user mobility).
- **VTP (VLAN Trunking Protocol):** Cisco proprietary protocol to synchronize VLAN configurations across a domain.

4.2 IEEE 802.11 (WLAN) MAC Protocols

- **Standard:** IEEE 802.11 (Wi-Fi) covers the Physical and MAC layers.
- **MAC Layer:** Uses **CSMA/CA (Collision Avoidance)** because collision detection (CSMA/CD) is unreliable in wireless networks.
- **Hidden Node Problem:** Solved using the optional **RTS/CTS (Request to Send/Clear to Send)** handshake to reserve the medium.
- **Coordination:**
 - **DCF (Distributed Coordination Function):** Contention-based (CSMA/CA).
 - **PCF (Point Coordination Function):** Optional centralized, contention-free polling by the Access Point (AP).

5 Network Applications and Protocols

5.1 Application Architectures

- **Client-Server:** Centralized (server: always-on, permanent IP, well-known ports).

- **Peer-to-Peer (P2P):** Decentralized (hosts act as both client/server). Highly scalable as peers contribute capacity.

5.2 Application Protocols

- **HTTP:** Stateless, uses TCP (Port 80). Uses **Cookies** for session management.
- **FTP:** Uses two TCP connections (Control: Port 21, Data: Port 20).
- **E-mail:** Uses SMTP (push/MTA, Port 25), POP3/IMAP4 (pull/MAA, Ports 110/143).
- **DNS:** Distributed database to map hostnames $\leftrightarrow$ IP addresses (UDP/TCP Port 53).

6 Internet-of-Things (IoT) Protocols

6.1 IoT Protocol Needs

- **Challenge:** Traditional Internet protocols (e.g., HTTP/TCP) are too "heavy" (high overhead, inefficient encoding) for resource-constrained IoT devices.
- **Thing:** Any Internet-addressable device that can sense, actuate, or compute in the Cyber-Physical Domain.

6.2 Constrained Application Protocol (CoAP)

- **Type:** Lightweight **web transfer protocol** (RESTful, equivalent to fast HTTP).
- **Transport:** Runs over **UDP** (may use DTLS for security).
- **Reliability:** Handled at the application layer using **Confirmable (CON)** messages, which require an **ACK**.

6.3 Message Queuing Telemetry Transport (MQTT)

- **Type:** **Publish/Subscribe** messaging protocol.
- **Architecture:** Centralized **Broker** mediates communication between clients (publishers/subscribers).
- **Transport:** Runs over **TCP/IP**.
- **Quality of Service (QoS):** Provides three reliability levels on top of TCP:
 - QoS 0: At-most-once (best effort).
 - QoS 1: At-least-once (acknowledged, duplicates possible).
 - QoS 2: Exactly-once (highest guarantee).

7 IP Layer and Routing Protocols

7.1 IP Fragmentation and ICMP

- **Fragmentation:** Occurs due to the physical link's **MTU** (Maximum Transmission Unit) limit. Reassembly occurs **only at the final destination**.
- **ICMP (Internet Control Message Protocol):** A network layer protocol that uses IP directly to report errors (e.g., Destination Unreachable, Time Exceeded) back to the original source.
- **Traceroute:** Uses progressively increasing **TTL** fields in IP packets to force routers to send back ICMP Time Exceeded messages, thereby mapping the route.

7.2 OSPF (Open Shortest Path First)

- **Type:** An **intradomain** link state routing protocol (within an AS).
- **Algorithm:** Each router computes the shortest path using **Dijkstra's algorithm** based on a complete network topology map.
- **Link State Advertisements (LSAs):** Packets containing link information (links, costs) that are reliably flooded to all routers in the area, ensuring an identical topological database.

8 Network Security Fundamentals

8.1 Security Services

- **Confidentiality:** Ensures message secrecy (using Symmetric/Asymmetric Encryption).
- **Message Integrity:** Verifies message has not been altered (using a Hash function to create a Message Digest).
- **Message Authentication:** Verifies the sender's identity (using Digital Signatures or MACs).
- **Non-repudiation:** Prevents the sender from denying the message was sent (requires a Digital Signature).

8.2 Security Protocols

- **Cryptography:** Uses symmetric keys (e.g., AES, for bulk encryption) and asymmetric keys (e.g., RSA, for digital signatures and key exchange).
- **SSL/TLS:** Provides transport layer security (encryption, authentication) for TCP.

- **IPsec:** Provides network layer security. **Tunnel Mode** is used by VPNs to encapsulate and secure the original IP packet.

9 IoT Security and Cyber-Physical Challenges

9.1 Traditional vs. Cyber-Physical Security

- **Traditional Security Approaches:** Great at protecting data in regard to identities or similar static attributes (e.g., Server or Client Authentication using certificates/ID/PW).
- **Cyber-Physical Security Challenge:**
 - Data relies more on physical relationships among parties than identities alone.
 - Identity-based (crypto) solutions alone do not prove relative physical context (e.g., "Is Car M really behind Car A?" or confirming a pizza delivery boy's location).
 - Cannot rely on GPS information due to low accuracy and spoofing risk.

9.2 IoT Security Challenges

- **Data Volume and Type:** Sensitive data creates greater risk of:
 - Data and identity theft.
 - Device manipulation.
 - Data falsification.
 - IP theft, server/network manipulation (especially due to data consolidation and analytics at the network edge).
- **Encryption and Key Exchange:**
 - Need for Transport and Application layer encryption (TLS and DTLS).
 - Need for secure key exchange algorithms over unsecured channels.
- **Authentication and Key Management:**
 - IoT devices and servers need to check each other's identity.
 - Challenges for provisioning services and tamper-resistant storage of keys (Asymmetric Public/Private keypairs vs. Symmetric keys).
- **Authorisation:** Determining Who Can Do What.
 - Simple form: Access Control Lists (ACLs).

- More general rules with complex conditions.
- **Capability based security:** A communicable and unforgeable token of authority associated with a set of access rights.
- IETF Work: **ACE** (Access Control in Constrained Environments) and **JOSE** (JavaScript Object Signing and Encryption).
- **Privacy:** The IoT has the potential to provide huge, unprecedented amounts of personal information that may last indefinitely.
 - Risk of abuse by individuals, criminals, companies, and governments.
 - **Strongly identifying information:** Address, date of birth, sexual orientation, etc.
 - **Weakly identifying information:** Can uniquely characterize a user when combined.
 - Companies need **privacy policies**, adherence to the **principle of data minimisation**, and regulatory compliance.
- **Legal Concerns:** Key questions relate to data scrubbing, maintenance, ease of unmasking/aggregation, data location, and ownership.

9.3 IoT Data Formats: JSON vs. CBOR

9.3.1 Data Format History

- Ad Hoc, Database Model, Document Model, Programming Language Model.

9.3.2 JSON (JavaScript Object Notation)

- **Model:** Programming Language Model (minimal, textual, subset of JavaScript).
- **Application Usage:** No schema needed for parsing; representation types include bool, number, string, null, map, array.
- **Extensibility:** Good, as parsers can ignore unknown keys in a key/value structure.
- **Limitations:**
 - No binary data (requires Base64 encoding).
 - Numbers are in decimal (requires parsing).
 - Format requires copying and escaping for strings.
 - No extensibility for types like date formats, leading to interoperability issues (e.g., I-JSON, RFC 7493).

9.3.3 CBOR (Concise Binary Object Representation)

- **Why a new format?** Designed for constrained node networks, prioritizing implementation compactness.
- **Key Design Goals:**
 1. Unambiguously encode most common data formats (JSON-like data).
 2. Compact implementation possible for encoder and decoder.
 3. Able to parse without a schema description.
 4. Serialization reasonably compact (but secondary to implementation compactness).
 5. Applicable to both constrained nodes and high-volume applications.
 6. Support all JSON data types, with conversion to and from JSON.
 7. Extensible (extended data can be parsed by earlier parsers).

9.4 Application Layer Security Protocols for IoT

- **Goal:** Achieve end-to-end security through intermediaries, across different transport layers (e.g., TCP, SMS, UDP, BLE), and for cross-protocol communication (e.g., HTTP client to CoAP server).
- **IETF Standardizations (Suitable for Constrained Environments):**
 - **COSE:** CBOR Object Signature and Encryption. A secure message format based on CBOR.
 - **OSCOAP:** Object Security for CoAP. Provides Authentication, encryption, integrity, and replay protection for CoAP.
 - **ACE:** Authentication and Authorization in Constrained Environments. A lightweight authorization framework (version of OAuth 2.0).
 - **EDHOC:** Key exchange protocol for establishing session keys with perfect forward secrecy, messages embedded as CBOR and COSE.

9.4.1 DTLS (Datagram Transport Layer Security)

- **Motivation:** TLS is incompatible with unreliable channels (like UDP) due to message loss. DTLS provides secured UDP communication.
- **Challenge:** Due to the connectionless nature of the UDP protocol, DTLS is vulnerable to Denial of Service (DoS) attacks with spoofed IP addresses.

- **Mitigation:** Uses a **Cookie** mechanism (Flight 1: ClientHello → HelloVerifyRequest → ClientHello*).

9.4.2 OSCOAP (Object Security for CoAP)

- **Function:** A security extension to CoAP, defined in terms of CBOR and COSE.
- **Security Services:** Provides end-to-end confidentiality, integrity, and replay protection for CoAP.
- **Mechanism:** Protects CoAP messages using the COSE_Encrypt0 format, which employs **Authenticated Encryption with Additional Data (AEAD)**. AES-CCM with 128 bit key/64 bit MAC is mandatory.
- **Benefits:** Is lightweight in terms of message overhead, RAM, and ROM compared to DTLS/TLS.

10 Multimedia Protocols and SIP for VoIP

10.1 Streaming Audio/Video

10.1.1 Types of Streaming

- **Streaming Stored Audio/Video:** Unicast and on-demand.
- **Streaming Live Audio/Video:** Multicast and live (e.g., broadcasting).
- **Interactive Audio/Video:** Real-time, bi-directional communication (e.g., VoIP, Video Conferencing, Cloud Games).

10.1.2 Streaming Architectures

- **Streaming using a Web Server:** Browser requests media file (GET), server responds, file is downloaded, media player runs.
- **Streaming using a Web Server with a Metafile:** Browser requests a metafile (e.g., SDP, XML format), media player receives the metafile and then initiates a second GET for the actual audio/video file.
- **Streaming using a Media Server and RTSP:** The media data is requested from a specialized media server rather than the web server.

10.1.3 Jitter Management

- **Jitter:** The variation in delay between packets in real-time data flow.

- **Solution:** To hide/prevent jitter, packets are **timestamped**, and a **Client side Playback Buffer** is used to separate the arrival time from the playback time. Sequence numbers are also added to packets.

10.2 Real-time Transport and Control Protocols

- **UDP Suitability:** UDP is more suitable than TCP for interactive traffic as it avoids retransmission delays, though it is unreliable.
- **RTP (Real-time Transport Protocol):**
 - Works with UDP to handle real-time traffic (makes up for UDP's deficiencies).
 - Main contributions: **time stamping**, **sequencing**, and **mixing facilities**.
 - Uses a temporary even-numbered UDP port.
- **RTCP (Real-time Transport Control Protocol):** Companion protocol to RTP that enables the sender to monitor the network condition and react (Quality of Service).
- **RTSP (Real Time Streaming Protocol):** An Application-layer **signaling** protocol (often over TCP) for media session control.
 - **Methods:** SETUP, PLAY, PAUSE, TEARDOWN, OPTIONS, DESCRIBE, RECORD, ANNOUNCE, REDIRECT, GET_PARAMETER, SET_PARAMETER.
- **SDP (Session Description Protocol):** Used for describing multimedia sessions for session announcement, invitation, etc.
 - **Includes:** Type of media (video, audio), transport protocol (RTP/UDP/IP), format of the media (PCMU audio, H.261 video), and information to receive the media (addresses, ports).

10.3 Session Initiation Protocol (SIP) for VoIP

- **Definition:** An application-layer control (signaling) protocol (IETF RFC 3261) for creating, modifying, and terminating sessions.
- **Characteristics:** Can be used for voice, video, instant messaging, etc. Follows HTTP style (text-based messaging) and uses URIs (e.g., `sip:bob@fhda.edu`).
- **Architecture:** SIP is an application-layer protocol that works with SDP and codecs, often running over TCP or UDP, and uses DNS for service records (DNS_{SRV}).
- **Components:**
 - **User Agents:** Clients (make requests) and Servers (accept requests).

- **Server Types:** Redirect Server, Proxy Server, Registrar Server, Location Server.
- **Registrar:** Accepts REGISTER requests from clients and places the information (IP address) into a location service for the domain it handles.
- **Session Flow (SIP Trapezoid):** Involves an Originating User Agent, Outgoing Proxy, DNS Server, Incoming Proxy, Location Server, Registrar, and a Terminating User Agent, with the media (RTP) being peer-to-peer.

10.3.1 SIP Messages (Methods)

- **INVITE:** Requests a session.
- **ACK:** Final response to the INVITE.
- **BYE:** Terminates a session.
- **REGISTER:** Sends user's address (IP) to the server.
- **OPTIONS:** Asks for server capabilities.
- **CANCEL:** Cancels a pending request.

10.3.2 SIP Responses

- **1XX** (Provisional): e.g., 100 Trying, 180 Ringing.
- **2XX** (Successful): e.g., 200 OK (Answers).
- **3XX** (Redirection): e.g., 302 Moved Temporarily.
- **4XX** (Client Error): e.g., 404 Not Found.
- **5XX** (Server Error): e.g., 504 Server Time-out.
- **6XX** (Global Failure): e.g., 603 Decline.

10.4 H.323 Protocol Suite

- **Overview:** ITU-T Recommendation covering technical requirements for multimedia communication in a Packet Based Network (originally for Videoconferencing).
- **Components:**
 - **H.323 Terminals:** Endpoints for real-time voice or video.
 - **Gatekeepers:** Central managers of VoIP services; mandatory functions include address resolution and authentication (similar to SIP Registrars).

- **Gateways:** Enable intercommunication between IP networks and legacy Switched Circuit Networks (SCNs) like ISDN and PSTN.
- **Protocol Suite:** Uses H.225 (Control), Q.931 (Setup), H.245 (Compression negotiation) (all over TCP), and RTP/RTCP (over UDP) for media.
- **SIP vs. H.323:** SIP is newer and offers far lower complexity, richer extensibility, and better scalability, which has led to a rapid increase in its use in Telco networks (especially with IP Multimedia Subsystem - IMS).

11 Software Defined Networking (SDN)

11.1 SDN Concept

- **Core Principle:** Separate the **Control Plane** and the **Data Plane** entities.
- **Control Plane (Smart):** Network intelligence and state are **logically centralized** and implemented as software on general-purpose hardware.
- **Data Plane (Dumb, Fast):** Forwarding logic is simple (e.g., switches/forwarding fabric) and abstracted from applications.
- **Benefit of Separation:** Independent evolution of software (control) and hardware (data), control behavior using higher-order programs, easier debugging/checking.
- **Benefit of Centralization:** Centralized decisions are easier to make (e.g., contrast distributed OSPF's complexity with a centralized controller running a simpler shortest path algorithm).

11.2 Problems with Traditional Networking

- **Complicated Core:** Routers run complex functions baked into the infrastructure (OSPF, BGP, Multicast, NAT, Firewalls, MPLS) resulting in a tightly coupled Management, Control, and Data plane.
- **Consequences of Complexity:** Buggy software, cascading failures, high operating costs, and slow evolution.
- **Closed Equipment:** Software is bundled with hardware (Vendor-specific interfaces), leading to long delays for new features and limited innovation.
- **Lack of Abstraction:** While data plane (layers) is well-abstracted, there are no powerful **control plane abstractions**.

11.3 OpenFlow Protocol

- **Definition:** Communication **interface/protocol** between the control and data plane of an SDN architecture.
- **Mechanism:** Allows direct access to and manipulation of the forwarding plane (flow table) of network devices over a **secure channel** (TLS).
- **Flow-Based Control:** Controller manages traffic flows by manipulating entries in the **Flow Table** in the switch.
- **Flow Table Structure:** Each entry has three fields:
 1. **Rule (Match):** Packet header fields to match (e.g., Switch port, MAC src/dst, Eth type, VLAN ID, IP src/dst, TCP psrc/pdst).
 2. **Action:** Operation to perform (e.g., Forward packet to port(s), Encapsulate and forward to controller, Drop packet, Send to normal processing pipeline).
 3. **Stats:** Packet and byte counters.
- **Packet Processing Flow:**
 1. Packet arrives at switch.
 2. Header is matched against Flow Table entries.
 3. If matched, perform action(s) and update counters.
 4. If no match (Table-miss), the Switch asks the controller by sending a message with the packet header.
- **OpenFlow Switch Components:** Datapath (Flow Table Pipeline, Group Table, Meter Table) and Control Channel (OpenFlow Channel).

11.4 Network Virtualisation (An SDN Application)

- **Concept:** Abstraction between the physical resources and their logical representation (similar to Server Virtualisation).
- **Function:** Makes a physical network appear as multiple logical ones (**network slicing**).
- **Goal:** Allows heterogeneous virtual networks that are isolated and independently managed to coexist over a shared physical network infrastructure.

11.5 SDN Lab Tools Overview

- **Mininet (Network Emulator):** Creates a realistic (real-time) virtual OpenFlow network (controller, switches, hosts, links) on a single real or virtual machine. Pro-

vides the Data Plane.

- **VirtualBox:** Cross-platform virtualization application used to host the Mininet VM.
- **Floodlight (SDN Controller):** An SDN Controller written in Java that runs within a JVM. Provides the Network Control Plane.
- **Procedure (Connecting Mininet to Controller):** `sudo mn ~controller=remote,ip=controll`